

Results

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Setup

Imports

```
## std lib
from pathlib import Path
from itertools import product

## 3rd p lib
from IPython.display import display, HTML
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.animation as animation

# project code
from src.config_parser import load_config
from src.viz import viz_sim, plot_loss_curves
```

```
experiment = "Big_World"
```

Loss Curves

```
sensing_terminals = ["Name", "Deitic", "Base"]
path_end = "loss_results.npz"
result_path = Path("results")
```

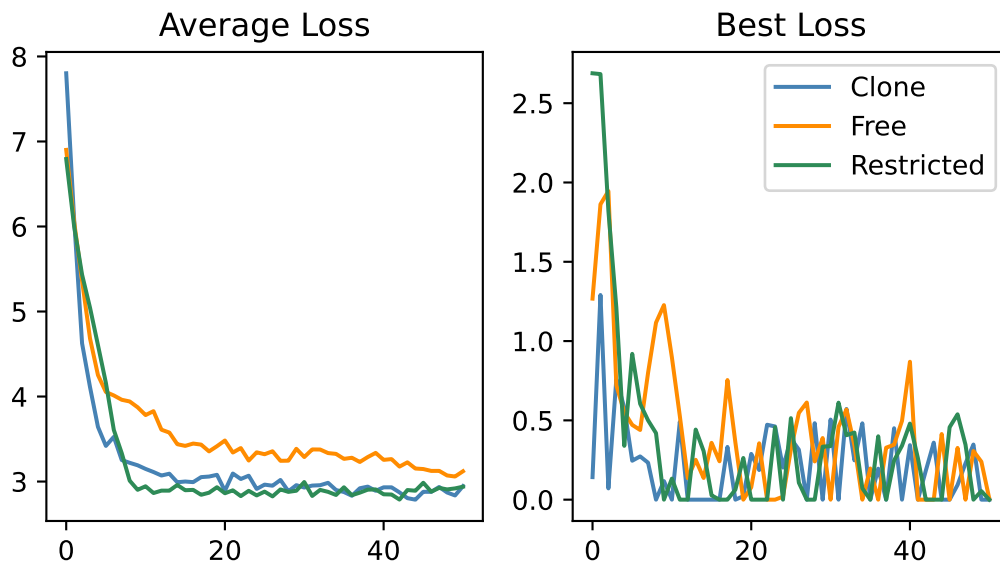
```

for terminal in sensing_terminals:
    clone_data = np.load(result_path / experiment / terminal / "Clone" / path_end)
    free_data = np.load(result_path / experiment / terminal / "Free" / path_end)
    restricted_data = np.load(result_path / experiment / terminal / "Restricted" / path_end)

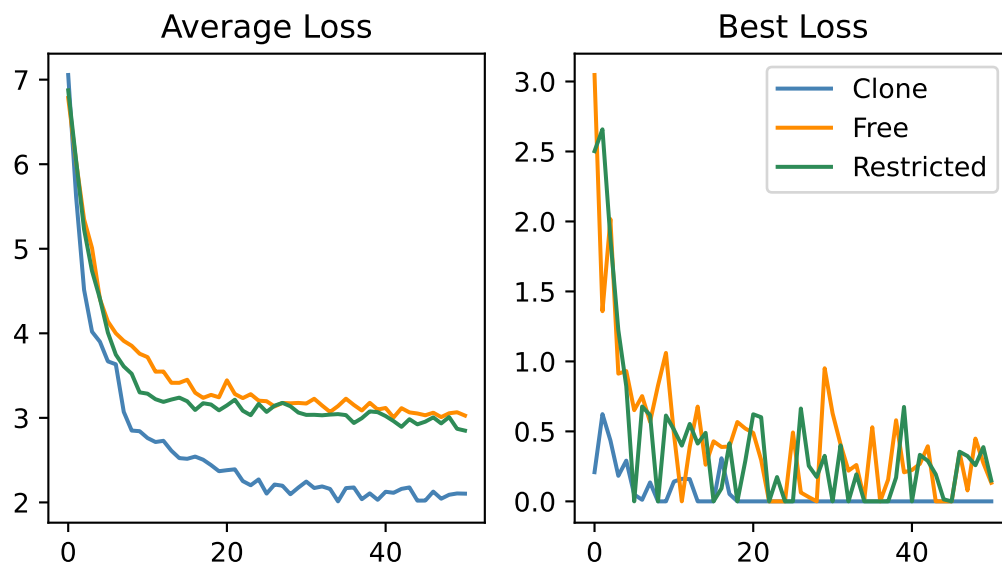
    plot_loss_curves(
        clone_data, free_data, restricted_data, title=f"{experiment}: {terminal}"
    )

```

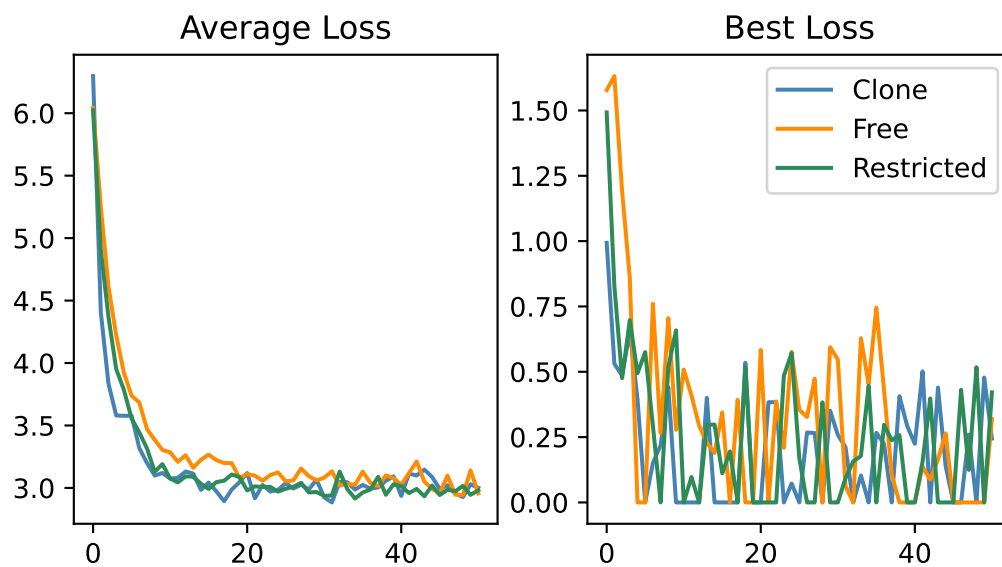
Big_World: Name



Big_World: Deitic



Big_World: Base



Animations

```
anim = Path("animations")
terminals = ["Base", "Deitic", "Name"]
strategies = ["Clone", "Free", "Restricted"]
qualities = ["best_positions", "random_positions"]
generations = ["first", "final"]

for terminal, strategy, quality, generation in product(
    terminals, strategies, qualities, generations
):
    path = anim / experiment / terminal / strategy / f"{quality}_{generation}.html"
    with open(path) as f:
        display(HTML(f.read()))
```

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